
Cooperative Emergence in Multi-Agent Stag-Hunt: A Comparison of A2C, PPO and MAPPO

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Autonomous and Adaptive Systems, A.A. 2025/2026
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Abstract

I study the emergence of cooperative behavior in the Gymnasium Stag-Hunt benchmark, a set of three grid-based stochastic games (Hunt, Harvest, and Escalation) that model different cooperation structures. Two independent agents are trained with A2C, PPO and MAPPO under standard, robust (domain randomization), and adaptive (curriculum) regimes. Results reveal a consistent symmetry between algorithm and environment: A2C reliably discovers a passive exploit on Hunt (avg. reward 2394, 100% coop) and dominates Harvest through timing precision (1743), while PPO partially discovers the same Hunt exploit but with extreme instability ($\sigma = 3324$). PPO excels on Escalation (rollout horizon, 66.68) where A2C converges to active defection (0.26, below the random baseline of 2.33). MAPPO converges to the risk-dominant foraging equilibrium on Hunt and does not systematically outperform decentralized baselines. The update mechanism is therefore as important a design choice as the network architecture.

1 Introduction

Multi-agent reinforcement learning (MARL) poses challenges beyond single-agent RL: agents must coordinate without explicit communication in environments that are non-stationary from each agent’s perspective (7). The Stag-Hunt game, a classic coordination dilemma from game theory, captures this tension: mutual cooperation yields the highest reward, but cooperating unilaterally leads to punishment, making the risk-dominant solo strategy a competing equilibrium.

The Gymnasium Stag-Hunt benchmark (2) instantiates this dilemma in three grid-based environments with qualitatively different cooperation structures: Hunt (simultaneous capture), Harvest (temporal synchronization), and Escalation (persistent commitment under partial observability). I use this variation to test whether different RL algorithms discover cooperative equilibria, and under what conditions they fail to.

I train two independent agents using A2C (3), PPO (4), and MAPPO (6) and compare their performance across all three environments. The main contributions of this work are:

- A comparison of A2C, PPO, and MAPPO across three cooperative environments with structurally different cooperation demands.
- An empirical analysis showing that update frequency, rollout horizon, and joint value estimation each determine whether an algorithm succeeds or fails on a given environment and that no single algorithm is universally best.

2 Environments

2.1 Gymnasium Stag-Hunt

Gymnasium Stag-Hunt (2) is a Gymnasium port of the original Gym Stag-Hunt (1). It provides three two-agent stochastic grid games on a 5×5 grid. At each timestep, both agents choose from 5 discrete actions (up, down, left, right, stay). Goal and obstacle positions are randomized at episode start; agent starting positions are fixed. Each episode lasts up to 1000 timesteps. I use coordinate observations (`obs_type='coords'`) and enable `flip_obs=True` so that agent 2 receives a perspective-flipped observation, making both agents' observation spaces identical and their policies interchangeable.

Hunt. Agents must jointly reach a stag for a large cooperative reward; each agent can also forage a plant alone for a smaller safe reward. With `stag_follows=True` the stag moves toward the nearest agent, enabling a passive exploit where stationary agents wait for the stag. With `stag_follows=False` agents must actively coordinate.

Harvest. Multiple plants are distributed on the grid and transition stochastically from young to mature each step. When an agent harvests a *mature* plant, *both* agents receive `mature_reward`; harvesting a young plant rewards only the harvester with the smaller `young_reward`. Each plant's maturity flag is included in the observation, so agents can learn to seek out mature plants. The challenge is temporal: agents must reach mature plants before they die.

Escalation. A mark appears on the grid. When both agents occupy the mark simultaneously, each earns +1 and the mark moves to an adjacent cell, starting a streak. If one agent occupies the mark alone, it receives a penalty proportional to the current streak length while the other agent earns nothing. Crucially, the streak counter is *not* included in the observation (6-dimensional coordinate vector), making sustained commitment hard to learn from observations alone.

3 Methods

3.1 Independent Learners

I adopt the Independent Learners (IL) framework (7): each agent acts on its own local observation and treats the other agent as part of the environment. Both agents share the same network weights (`flip_obs=True` ensures observational symmetry) but maintain separate experience buffers, effectively training a single symmetric policy from both agents' perspectives.

3.2 Network Architecture

Actor and critic are separate two-layer MLPs:

$$\begin{aligned}\pi(a | s) &: \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(5, \text{softmax}), \\ V(s) &: \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(1, \text{linear}).\end{aligned}$$

All networks are implemented in TensorFlow/Keras with Adam optimizer.

3.3 A2C

Advantage Actor-Critic (3) updates parameters every $T=5$ steps. The n -step return is computed via backward recursion:

$$\hat{R}_t = r_t + \gamma \hat{R}_{t+1}, \quad \text{with } \hat{R}_T = \begin{cases} V(s_T) & \text{if not done} \\ 0 & \text{if done} \end{cases}$$

The advantage is $A_t = \hat{R}_t - V(s_t)$ and the combined loss is:

$$\mathcal{L} = \mathbb{E} \left[-\log \pi(a|s) \cdot A_t + c_1 (\hat{R}_t - V(s))^2 - c_2 H(\pi) \right]$$

with $c_1=0.5$, $c_2=0.1$, $\gamma=0.99$, $\text{lr}=10^{-3}$.

Gradients are clipped by global norm at 0.5 to prevent instability.

3.4 PPO

Proximal Policy Optimization (4) collects a rollout of $T=256$ steps and estimates advantages via GAE (5). Defining the TD residual $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, the advantage estimate is:

$$\hat{A}_t = \sum_{l=0}^{T-1-t} (\gamma\lambda)^l \delta_{t+l} = \delta_t + \gamma\lambda \hat{A}_{t+1}, \quad \hat{A}_T = 0,$$

with $\lambda=0.95$. Parameters are then optimized for $K=4$ epochs with the combined loss:

$$\mathcal{L} = \mathbb{E}[-\mathcal{L}_t^{\text{CLIP}} + c_1 \mathcal{L}_t^{\text{VF}} - c_2 H(\pi)],$$

where the clipped surrogate objective is:

$$\mathcal{L}_t^{\text{CLIP}} = \min\left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t\right),$$

and $\rho_t = \pi(a_t|s_t)/\pi_{\text{old}}(a_t|s_t)$ is the probability ratio. To prevent the value function from drifting too far from V_{old} across the K epochs, the critic loss is also clipped:

$$\mathcal{L}_t^{\text{VF}} = \max\left[\left(\hat{R}_t - V(s_t)\right)^2, \left(\hat{R}_t - \text{clip}(V(s_t), V_{\text{old}}(s_t)-\epsilon, V_{\text{old}}(s_t)+\epsilon)\right)^2\right],$$

where $\hat{R}_t$ is the bootstrapped n -step return (same recursion as A2C). Hyperparameters: $\epsilon = 0.2$, $\lambda = 0.95$, $\gamma = 0.99$, $c_1 = 0.5$, $c_2 = 0.01$, $\text{lr} = 3 \times 10^{-4}$. Gradients are clipped at norm 0.5 and the learning rate decays linearly to 10^{-6} over 10 000 updates.

3.5 MAPPO

MAPPO (6) extends PPO with a centralized critic that receives the concatenated observations of both agents $[s_1, s_2]$ during training (CTDE paradigm). At execution time each actor uses only its local observation. The actor update follows the standard PPO clipped objective; only the critic input changes. Note that with `obs_type='coords'` each agent's observation already contains the positions of both agents and all goals, so the centralized critic's global view provides limited additional information beyond what is locally observable.

3.6 Training Variants (Hunt)

Three variants are evaluated on Hunt: **Standard**, `stag_follows=True`; **Robust**, stochastic `stag_follows` (50% per episode, domain randomization); **Adaptive**, curriculum: warm-start from standard weights, then fine-tune with `stag_follows=False`. Harvest and Escalation use standard training only.

4 Experiments

4.1 Setup

All agents are trained for 5000 episodes using the hyperparameters in Table 1. Evaluation is performed over 100 episodes with stochastic policy sampling ($\pi(a|s)$ sampled, not `argmax`). Results report mean $\pm$ standard deviation over the 100 evaluation episodes. A random agent is included as a lower-bound baseline. All runs use seed 42.

4.2 Hunt

Standard. A2C reliably discovers a *passive exploit*: agents remain stationary while the stag approaches (241 stag catches/ep, avg. 2394, 100% coop). PPO partially discovers the same exploit, averaging 174 stag catches/ep and 72% cooperation, but is markedly unstable ($\sigma=3324$). This high variance suggests that PPO's 256-step rollout is sensitive to initial trajectory quality: episodes that happen to start near the passive exploit reinforce it, while others drift toward defection. MAPPO, by contrast, converges to the risk-dominant foraging equilibrium (43 plant/ep, only 6% coop), indicating that the centralized critic does not provide the stability needed to discover the passive strategy. With `stag_follows=False`, A2C (130) and PPO (139) reach comparable active stag-hunting performance, while MAPPO (302) retains its plant-based strategy.

Table 1: Hyperparameters used for all experiments.

Hyperparameter	A2C	PPO / MAPPO
Learning rate	10^{-3}	3×10^{-4}
Discount γ	0.99	0.99
Steps per update T	5	256
Epochs per rollout K	—	4
GAE λ	—	0.95
Clip range ϵ	—	0.2
Entropy coef c_2	0.1	0.01
Value coef c_1	0.5	0.5
Gradient clip norm	0.5	0.5

Table 2: Hunt evaluation. Coop% = fraction of episodes with at least one stag catch. stag=T/F refers to the stag_follows evaluation condition.

Algorithm	Variant	Avg Reward	Stag/ep	Plant/ep	Coop%
Random	—	-1629.74 ± 105	11.07	59.18	100%
A2C	standard (stag=T)	2394 ± 100	240.5	1.01	100%
A2C	standard (stag=F)	130 ± 35	13.39	0.91	100%
A2C	robust (stag=T)	2419 ± 94	242.0	0.23	100%
A2C	adaptive (stag=F)	576 ± 111	4.56	276.75	91%
PPO	standard (stag=T)	344 ± 3324	174.1	0.13	72%
PPO	standard (stag=F)	139 ± 36	14.01	0.25	100%
PPO	robust (stag=T)	-4683 ± 1478	10.23	0.00	4%
PPO	adaptive (stag=F)	634 ± 73	1.68	319.42	94%
MAPPO	standard (stag=T)	-21.89 ± 248	0.15	43.28	6%
MAPPO	standard (stag=F)	301.84 ± 77	11.19	115.66	100%
MAPPO	robust (stag=T)	-4985.00 ± 0	0.00	0.00	0%
MAPPO	adaptive (stag=F)	644.16 ± 52	2.11	317.66	99%

Robust. Domain randomization is catastrophic for both PPO (-4683) and MAPPO (-4985 , zero variance, policy collapsed to a single action). A2C robust (2419) matches standard performance, suggesting step-level updates are more resilient to non-stationarity than long-rollout methods.

Adaptive. Curriculum learning closes the gap: MAPPO adaptive (644) > PPO adaptive (634) > A2C adaptive (576). All three converge to a stable foraging equilibrium (276–319 plant/ep, $\geq 91\%$ coop). MAPPO adaptive is robust across both evaluation conditions (stag=T: 638, stag=F: 644), where the centralized critic helps maintain consistency during fine-tuning.

4.3 Harvest and Escalation

Harvest. All three algorithms achieve 100% cooperation (agents learn to target mature plants), but harvest throughput degrades: A2C (1743) $\gg$ PPO (815) > MAPPO (654). Since mature-plant events have a short temporal window (chance_to_die = 0.1 per step), A2C’s 5-step update captures the precise step that triggers the cooperative reward. PPO’s 256-step rollout dilutes this signal across many transitions, reducing timing precision. MAPPO’s centralized critic provides no advantage here; the challenge is temporal, not coordinative. Moreover, conditioning the critic on the concatenated observations of both agents introduces additional variance into the value estimate on a task where the cooperative signal is already precise and locally observable, which may explain MAPPO’s lower throughput relative to standard PPO.

Escalation. The ranking reverses completely: PPO (66.68) $\gg$ MAPPO (10.68) $\gg$ A2C (0.26). A2C converges to active defection (6% coop), scoring below the random baseline (2.33). The cooperation streak is absent from the 6-dimensional observation, requiring agents to maintain commitment over long horizons. PPO’s 256-step rollout captures this long-range credit assignment; A2C’s 5-step

Table 3: Harvest (left) and Escalation (right) evaluation. Coop% = episodes where both agents cooperate. Coop/ep = cooperative events per episode.

Harvest			
Algorithm	Avg Reward	Coop/ep	Coop%
Random	356.52 $\pm$ 34	67.88	100%
A2C	1743.42 $\pm$ 128	177.63	100%
PPO	814.79 $\pm$ 173	106.81	100%
MAPPO	654.18 $\pm$ 162	90.20	100%

Escalation			
Algorithm	Avg Reward	Coop/ep	Coop%
Random	2.33 $\pm$ 2.09	1.28	70%
A2C	0.26 $\pm$ 1.19	0.13	6%
MAPPO	10.68 $\pm$ 23.26	5.34	46%
PPO	66.68 $\pm$ 67.90	33.34	100%

horizon cannot. MAPPO is intermediate: the centralized critic improves spatial coordination toward the mark over A2C, but cannot substitute the long rollout for sustaining the streak.

4.4 Summary

Table 4: Algorithm ranking per environment (avg. reward, best variant).

	Hunt (stag=F)	Harvest	Escalation
1st	MAPPO adap. (644)	A2C (1743)	PPO (66.68)
2nd	PPO adap. (634)	PPO (815)	MAPPO (10.68)
3rd	A2C adap. (576)	MAPPO (654)	A2C (0.26)
Key factor	Curriculum	Update frequency	Rollout horizon

Which algorithm wins depends entirely on the environment. The determining factor is the structural match between the algorithm’s update mechanism and the environment’s cooperation requirement: step precision (Harvest), passive exploitation (Hunt standard), or long-horizon commitment (Escalation).

5 Conclusion

The results show that A2C, PPO, and MAPPO converge to distinct strategies across the three environments tested, and that which algorithm to use depends on the structure of the task not on general rankings. A2C’s frequent updates work better when cooperation requires precise timing, while PPO’s longer rollout horizon is necessary for sustained commitment under partial observability. The centralized critic in MAPPO adds robustness when training conditions are stochastic, as seen in the adaptive curriculum.

Limitations. All experiments use a single random seed, so reported variances reflect within-run stochasticity rather than across-seed variability. The CTDE paradigm is also constrained by the observation space: with `obs_type='coords'` the centralized critic receives little information beyond what each agent already observes independently, limiting the potential benefit of MAPPO over decentralized baselines.

Future work. An evolutionary hyperparameter search (8 individuals, genetic operators) was explored to optimize A2C coefficients automatically. A proof-of-concept run identified a promising region ($\text{lr} \approx 5.6 \times 10^{-3}$, $\text{entropy_coef} \approx 0.21$) after a single generation, but the required compute (~ 70 min per individual on CPU) made a full run intractable within available resources. Future work could run this search with GPU acceleration to assess whether tuned hyperparameters close the gap between algorithms on the harder environments.

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